

1 Performance Report

Overview

The analysis concerns workplace/ machine-related performance data for a certain period of time and a certain number of workplaces. The results depend on the selections made. The performances displayed in graphic form for quantities and duration provide the production controller with the desired information immediately at a glance.

Selection criteria

The application provides the following selection criteria:

Workplace

This selection criterion references the workplace in the machine or workplace master data. You can also run a search using wildcards (placeholders *).

Group

This selection criterion references the group in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected group. You can also run a search using wildcards.

Cost center

This selection criterion references the cost center defined in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected cost center. You can also run a search using wildcards.

Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group. You cannot run a search using wildcards.

Date from ... to (shift/ time)

The period for the data to be evaluated can be limited via the date selection option.

If a selection is made via a shift (shifts), the shift date is evaluated. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

Performance report

The performance report concerns workplace/ machine-related performance data for a certain period of time and a certain number of workplaces. The results depend on the selection and therefore on the selection criteria made available on the selection panel.

The following workplace/ machine-related master data is available:

- Workplace
- Short designation
- Group
- Yield (primary quantity unit)
- Scrap (primary quantity unit)
- Unit (primary quantity unit)
- $\text{Rate} = 100 / ((\text{yield (primary quantity)} + \text{scrap (primary quantity)} + \text{rework quantity (primary quantity)} + \text{open quantity (primary quantity)}) * \text{yield (primary quantity)})$
- $\text{Scrap rate} = 100 / ((\text{yield (primary quantity)} + \text{scrap (primary quantity)}) * \text{scrap (primary quantity)})$
- Number of booked cycles
- $\text{MUT} = \text{RPA11}$
- $\text{Downtimes} = \text{RPA01} + \text{RPA02} + \text{RPA03} + \text{RPA04} + \text{RPA05} + \text{RPA06} + \text{RPA07} + \text{RPA08} + \text{RPA09} + \text{RPA10}$
- $\text{Total} = \text{production} + \text{downtime}$

- Rate of capacity utilization

The rate of capacity utilization represents the ratios derived from the effective runtime and the machine operation time.

Rate of capacity utilization = $100 / (RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10 + RPA11) * RPA11$

- Assignment utilization rate:

The assignment utilization rate represents the relationship resulting from the effective runtime (main utilization time) and the adjusted machine operation time (i.e. without scheduled downtimes).

Assignment utilization rate = $100 / (\text{total of RPA 1 to RPA 11 minus RPA 6}) * RPA11$

- Technical efficiency

The variable showing the technical efficiency represents the sum total of the effective runtime and interruptions caused by technical (machine-related) disturbances. The times for all other disturbances (e.g. organization-related disturbances) are not accounted for in this calculation:

Technical efficiency = $100 / (RPA02 + RPA11) * RPA11$

Graphic efficiency report

This detail application creates two graphic presentations:

Shown in the resource performance accounts graphic are the posted durations broken down by each of the resource performance accounts 1-12 in the form of a pie chart. Illustrated in the legend to the right of the chart for each of the resource performance accounts are, in addition to the color, the RPA number and the designation, also the durations as absolute values as well as their share of the total number of resource performance accounts in percent.

Shown in the primary quantities graphic are the posted quantities broken down by each separate quantity types (yield, scrap, rework and open quantity), also in the form of a pie chart. Illustrated in the legend to the right of the chart for each of the resource performance accounts are, in addition to the color and the quantity type, also the quantities as absolute values as well as their share of the total number of resource performance accounts in percent.

Graphic performance profile

Two graphics are available in the graphic performance profile detail application:

Based on the entries selected in the table, the posted absolute time durations - time of production and standstill period - are shown in the Durations chart as stacked bars. The values (grid spacing on the X axis) are presented on the shift level.

Similar to the durations, the quantities of the posted absolute quantities (primary quantity unit) - yield, scrap - are also shown in the chart as stacked bars. The values (grid spacing on the X axis) are also presented on the shift level.

Graphic capacity utilization profile

Shown in a line chart in the graphic capacity utilization profile detail application are the key figures

- Rate of capacity utilization
- Assignment utilization rate
- Technical efficiency
- Rate

distributed over a period of time (grid spacing on the X axis on a shift level). The color code for each of the key figures is listed in the legend on the right side of the chart.

Please note: for technical reasons, the partial paths of the lines are shown in the form of a dotted line.